

Final Demo Script (Group 16)

Demo Goal

Show end-to-end operation on IEEE 14-bus:

1. Load case
2. Validate data
3. Build Ybus / view Zbus
4. Run power flow (GS and NR)
5. Run 3PH fault study at selected bus
6. Export selected outputs

Live Steps

1. Start the app.
 - macOS: `./Launch_PowerFlow_UI.command``
 - Windows: `WindowsLauncher_PowerFlow_UI.bat``
2. In the app:
 - Select ``IEEE 14-bus``
 - Click ``Load File``
3. Click ``Validate``.
 - Show that checks run (slack bus, numbering, branch/generator consistency).
4. Click ``Create Ybus``.
 - Show matrix tabs: ``Ybus`` and ``Zbus``.
5. Click ``Power Flow`` with ``NR``.
 - Point out convergence and iterations.
 - Show bus voltages, line flows/losses, and power balance.
6. Change solver to ``GS`` and run again.
 - Highlight convergence/iteration difference vs NR.
7. Configure fault panel:
 - Fault type: ``3PH``
 - Faulted element: ``Bus``
 - Faulted bus: ``4``
 - ``Rf = 0``, ``Xf = 0``
8. Click ``Fault Study``.
 - Show fault current table and post-fault voltages.
9. Click top ``Export``.
 - Select desired datasets with checkboxes.
 - Set export label.
 - Click bottom ``Export`` button to create timestamped folder under ``exports/``.

Backup CLI Demo (if UI issues occur)

```
```bash
python -m pytest -q
./scripts/run_final_demo.sh --fault-bus 4 --tolerance 1e-6 --max-iterations 200
```
```

Show:

- ``docs/validation/final_ieee14_powerflow_summary.csv``
- ``docs/validation/final_ieee14_fault_summary.csv``
- ``docs/validation/final_submission_summary.txt``